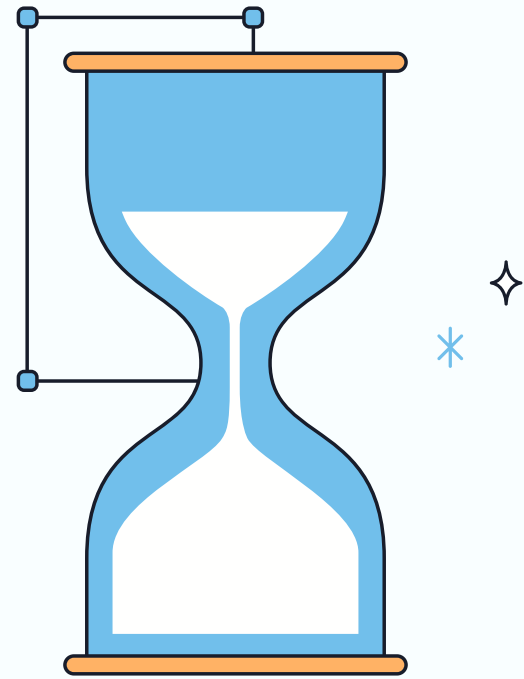




MM-BD

Post-Training Detection of Backdoor Attacks using Maximum Margin Statistics

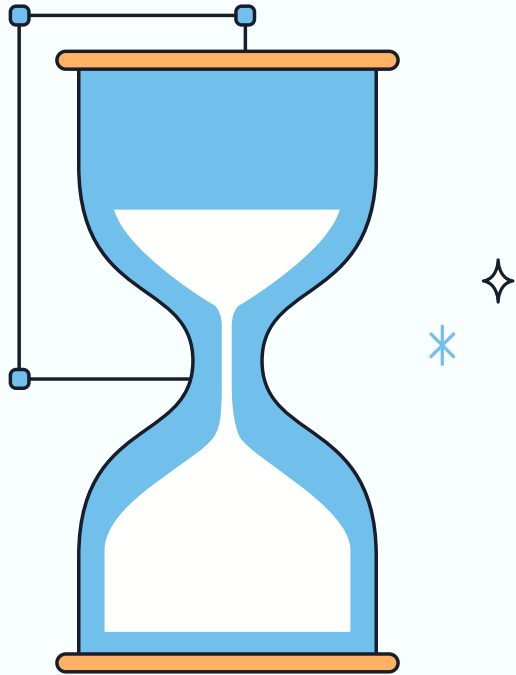


Researcher
Hang Wang



Date
May 9, 2026

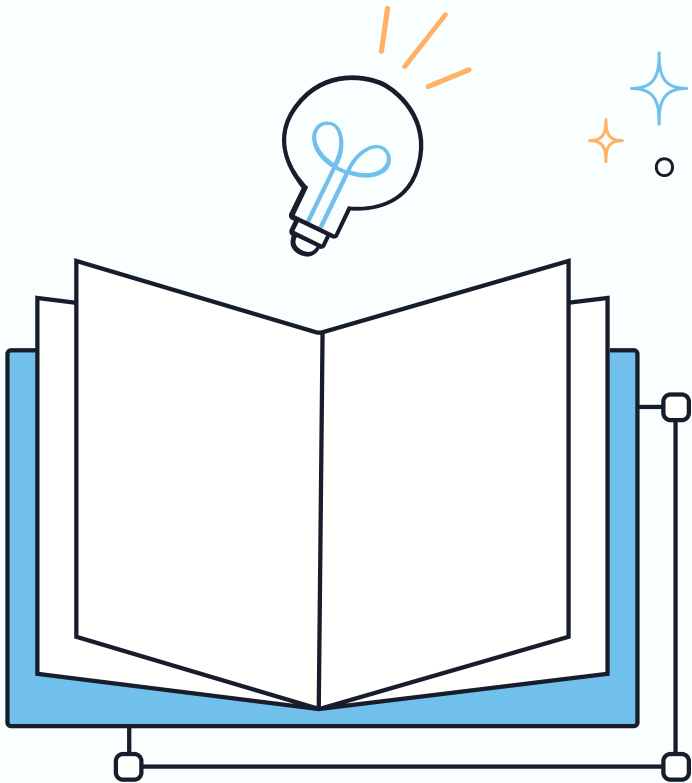




SUBMITTED BY

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Date
May 9, 2026



Introduction & Problem Statement

Deep Neural Networks are vulnerable to Backdoor (Trojan) Attacks.

In a backdoor attack:

- the model behaves normally on clean inputs
- but misclassifies inputs containing a hidden trigger

The attacker poisons the training data using trigger-embedded images labeled as a target class.

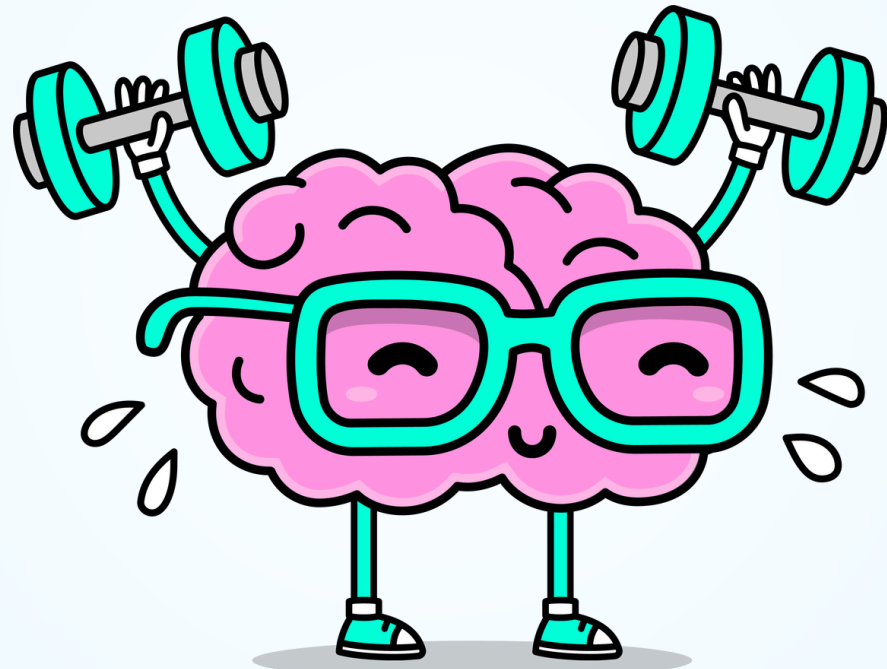
As a result:

- clean accuracy remains high
- malicious behavior activates only when the trigger is present

This makes backdoor attacks difficult to detect using normal validation accuracy.



Motivation of MM-BD



Main Problem

Most existing methods:

- assume a specific trigger type
- require clean training samples
- fail for arbitrary backdoor patterns
- struggle with multiple source classes

MM-BD aims to:

- detect arbitrary backdoor patterns
- work without clean training data
- detect attacks statistically using model logits

Backdoor Attack Mechanism

BadNet Patch Attack

→ This work uses the classical BadNet patch-based backdoor attack. A small trigger patch is inserted into images during training and testing.

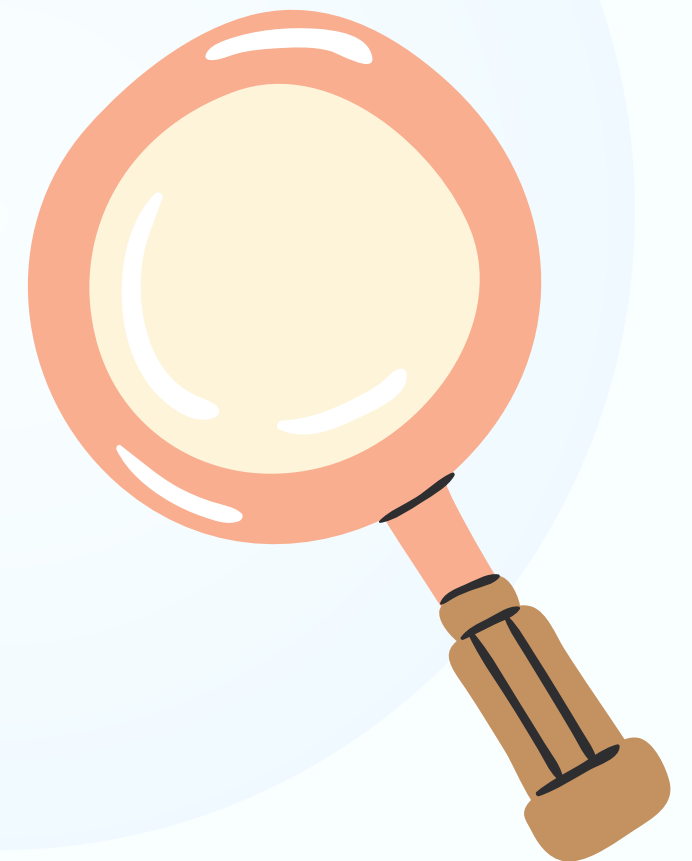
The model learns:

→ “If this trigger appears → predict target class.”

Key Property

→ A successful backdoor attack achieves:

- High Clean Accuracy (ACC)
 - High Attack Success Rate (ASR)
- while remaining visually stealthy.



Backdoor Attack Mechanism

Backdoor Trigger Visualisation — BadNet Patch (3×3, bottom-right)



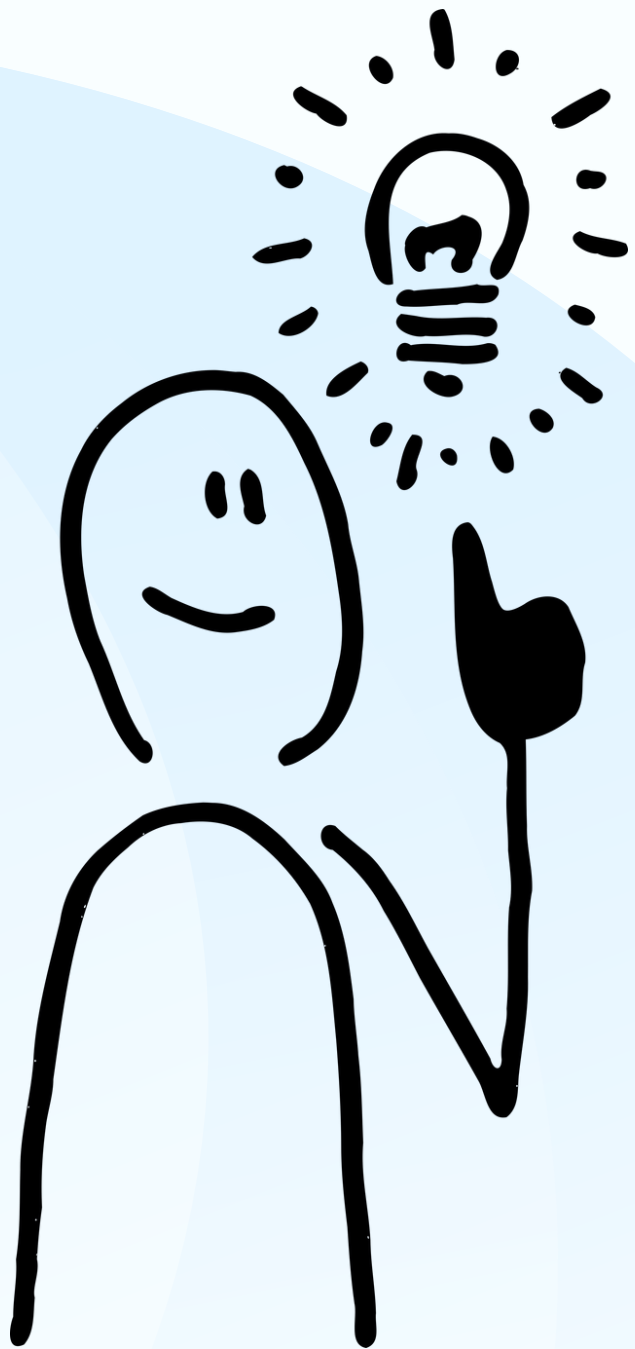
Backdoor Embedding Formula

$$\tilde{x} = (1 - m) \odot x + m \odot u$$

Where:

- x = clean image
- u = trigger pattern
- m = binary mask
- $\tilde{x}$ = poisoned image

Core Idea of MM-BD



Observation

Backdoor attacks create abnormal behavior in the model's logit space.

The target class develops an unusually large margin because:

- the trigger appears repeatedly during poisoning
- the network overfits to the trigger
- target logits become dominant

What Are Logits?



What are logits?

Before the final softmax layer, the model produces one raw score per class. Higher score = more confident. The margin = (target class score) – (highest competitor's score).

plane: 1.2

car: 0.8

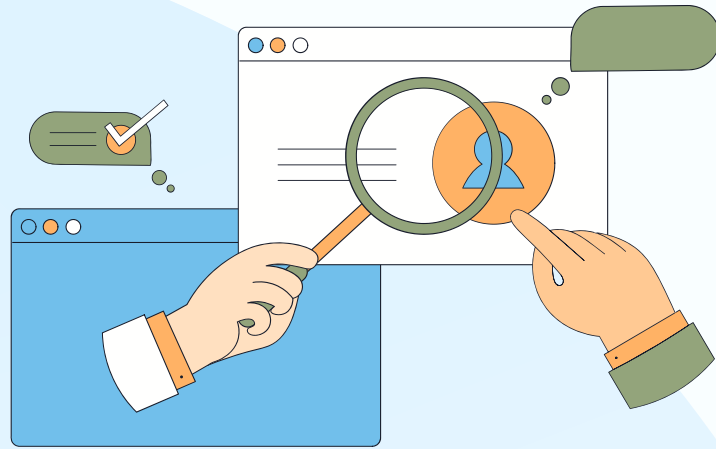
bird: 8.4
← target class

cat: 0.5

dog: 0.9

Margin
= 8.4 – 1.2 = 7.2

Core Idea of MM-BD



Maximum Margin Statistic

$$\max_{x \in X} (g_t(x) - \max_{k \neq t} g_k(x))$$

- Where:
- $g_t(x)$ = target class logit
- $g_k(x)$ = logits of other classes

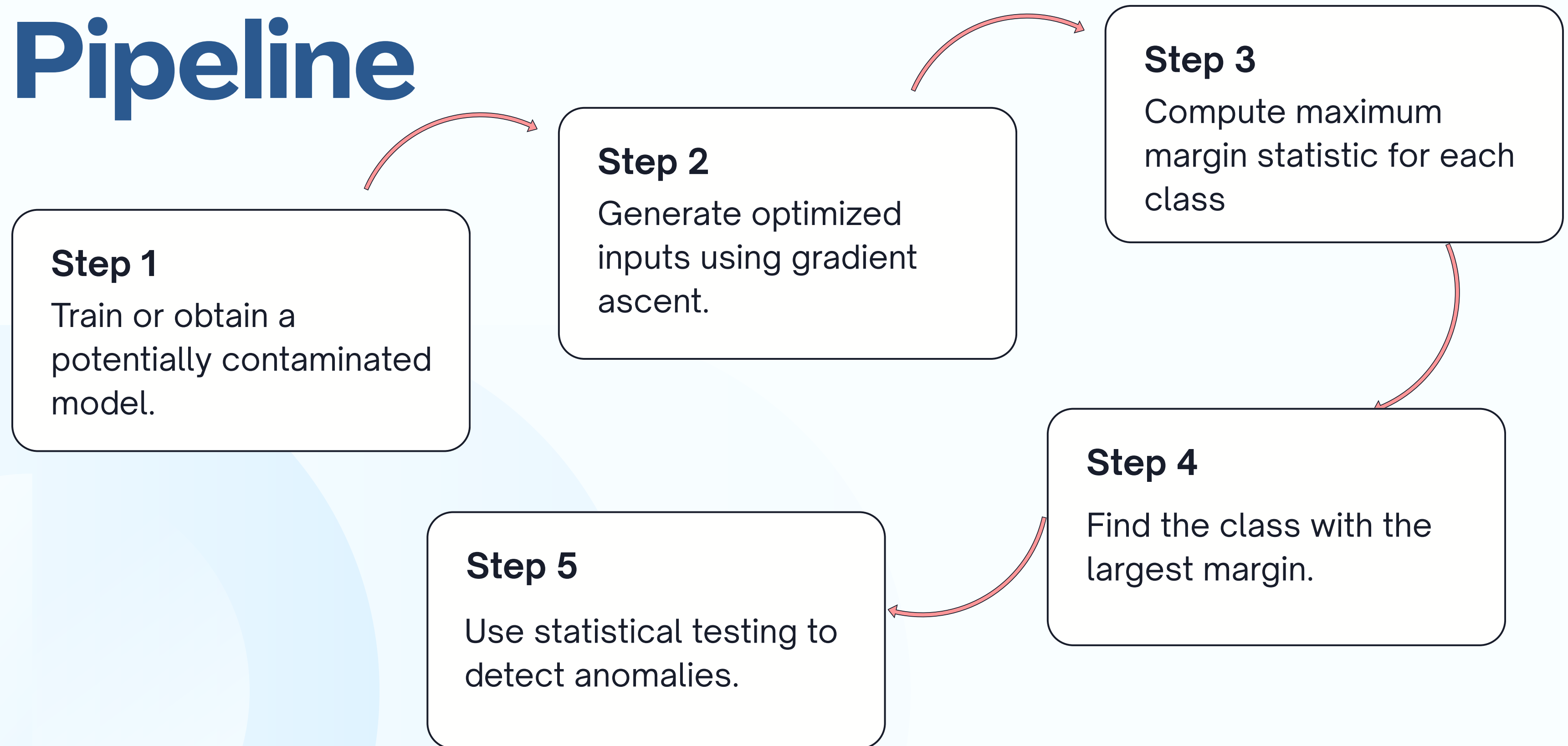
Key Insight

For a poisoned model:

- the target class margin becomes a statistical outlier
- this anomaly can be detected without knowing the trigger type

MM-BD Detection

Pipeline



P-Value Formula

$$pv = 1 - H_0(r_{max})^{K-1}$$

Where:



- rmax = largest margin statistic
- H0 = estimated null distribution
- K = number of classes

Decision Rule

If:

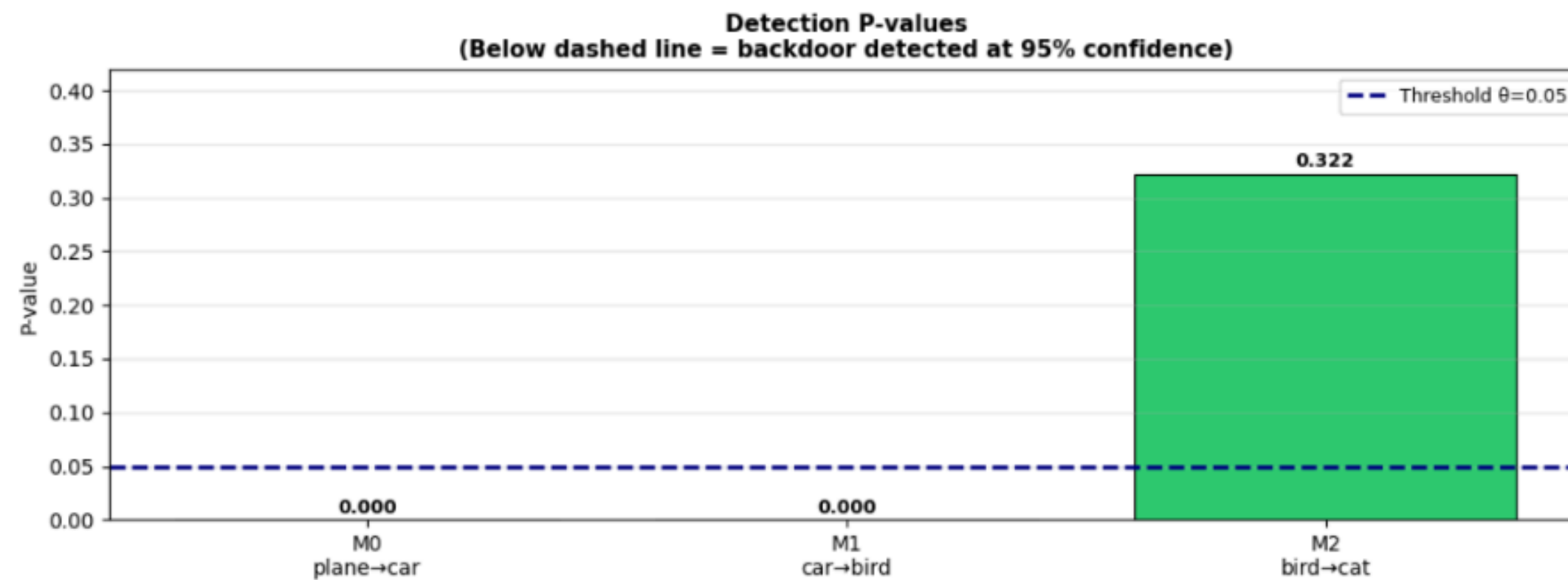


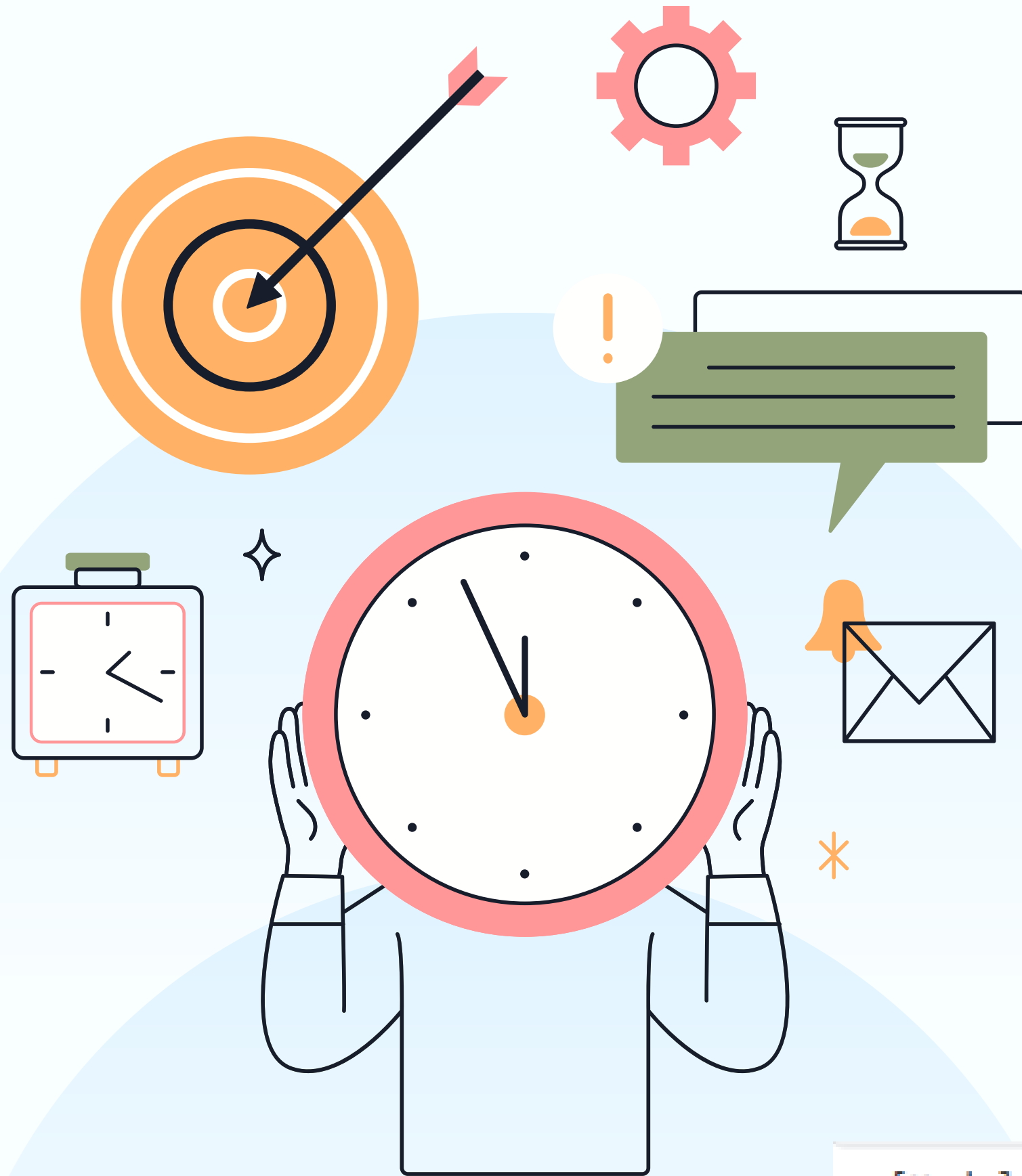
- $p < 0.05$

Then:

- backdoor attack is detected

P-Value Curve





Experimental Setup

Experimental Configuration

Dataset

- CIFAR-10

Architecture

- ResNet-18

Attack Type

- BadNet (3×3 patch trigger)

Training

- 3 contaminated models
- 20 epochs per model

Poisoned Samples

- 500 poisoned training samples
- 1000 poisoned test samples

[Model 0]	plane → car	train_poison= 500	test_poison=1000
[Model 1]	car → bird	train_poison= 500	test_poison=1000
[Model 2]	bird → cat	train_poison= 500	test_poison=1000

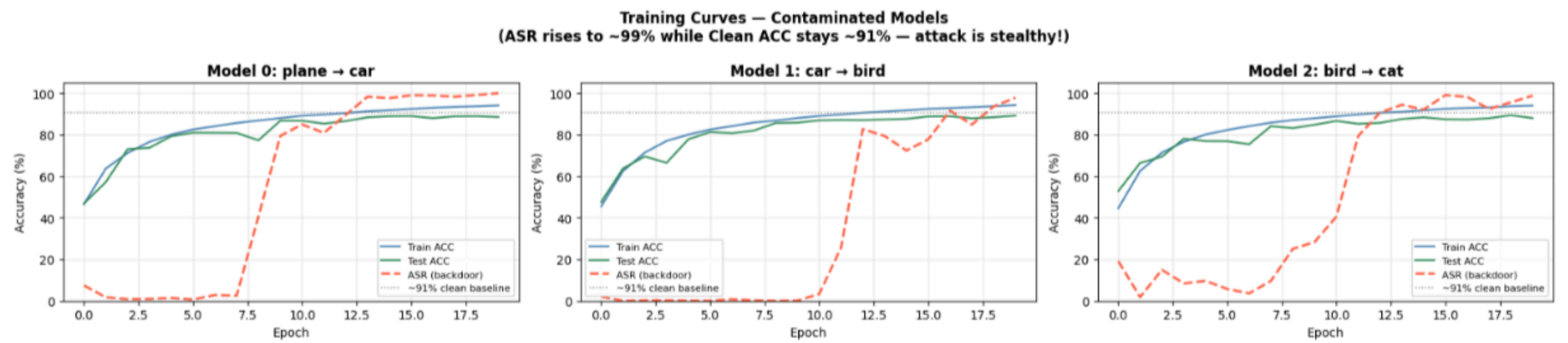
Training Results

Model Training Performance

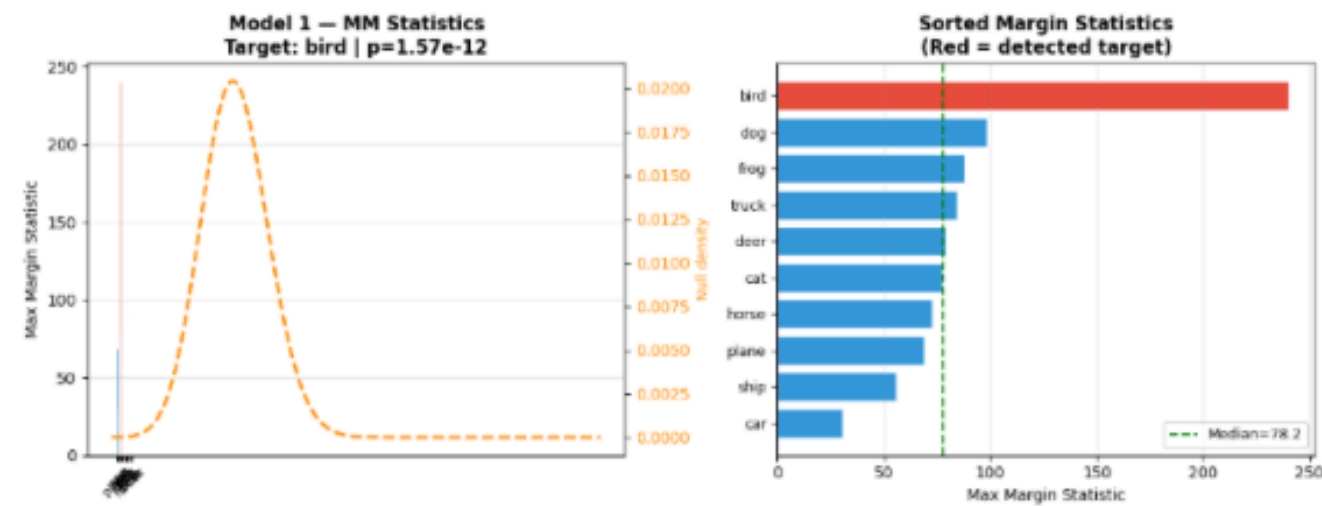
The contaminated models achieved:

- high clean accuracy
- very high attack success rate

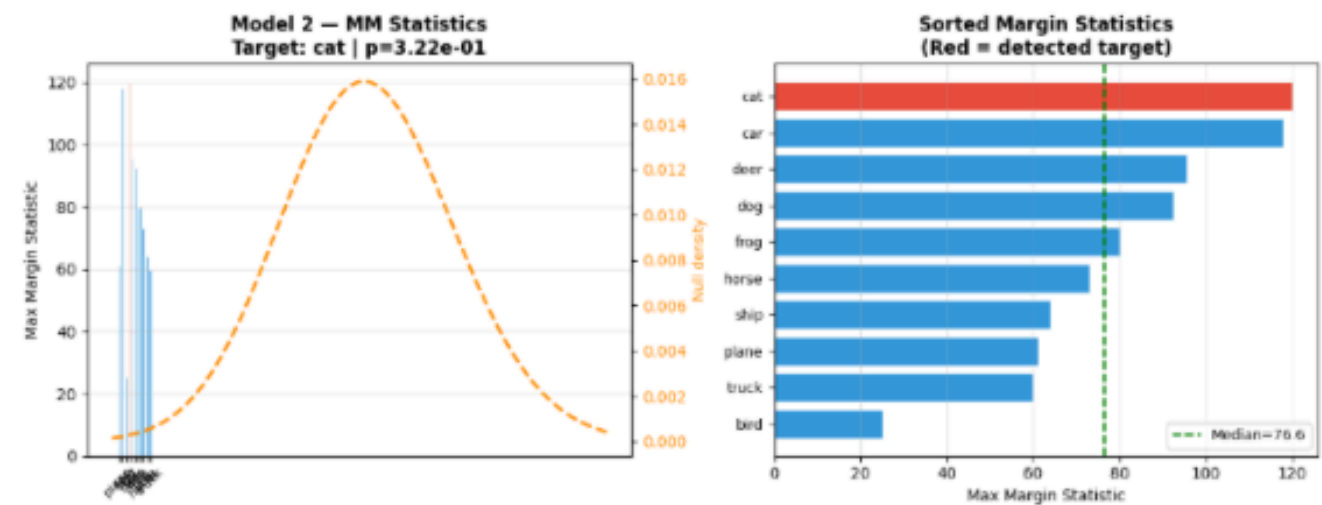
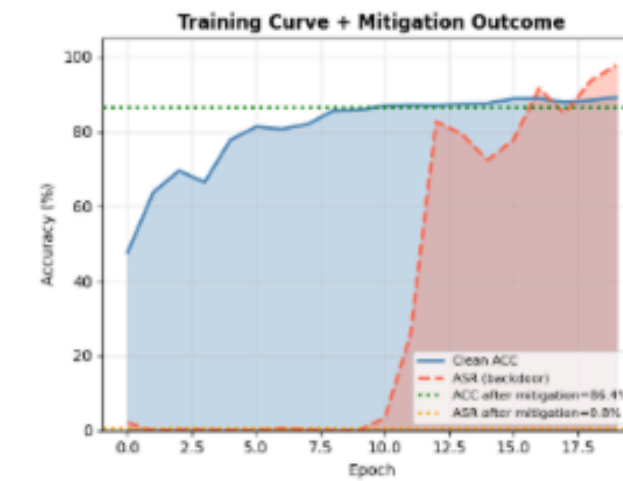
This indicates successful stealthy backdoor attacks.



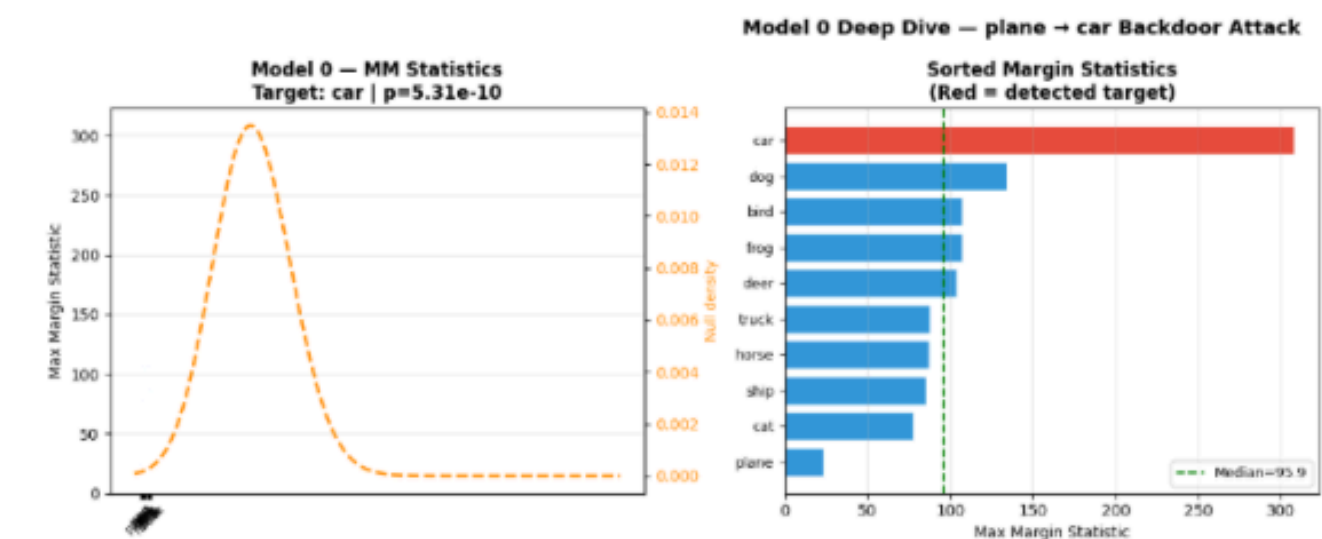
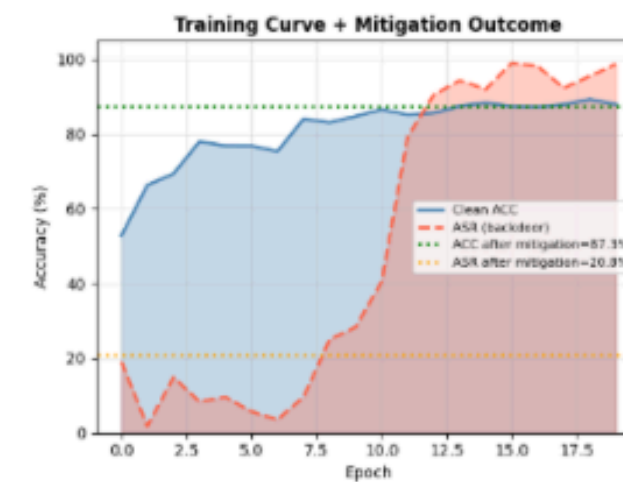
Detection Results



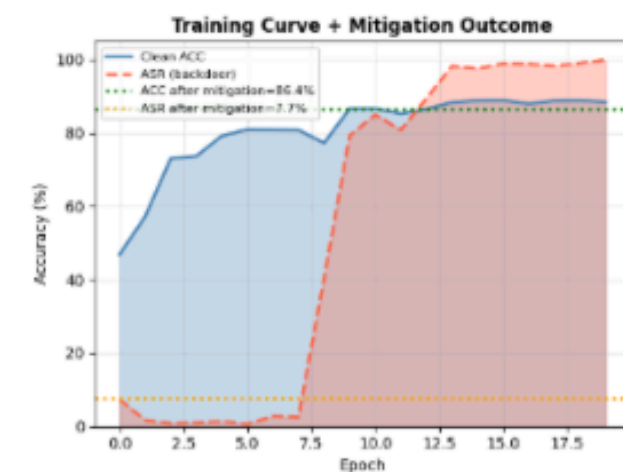
✓ Deep dive plot saved for Model 1



✓ Deep dive plot saved for Model 2



✓ Deep dive plot saved for Model 0



Analysis

Analysis

- Models 0 and 1 showed extremely low p-values.
- Their target classes became strong statistical outliers.
- Model 2 was not detected because the margin anomaly was weaker.

Possible reasons:

- only 20 epochs used
- insufficient convergence
- weaker target separation

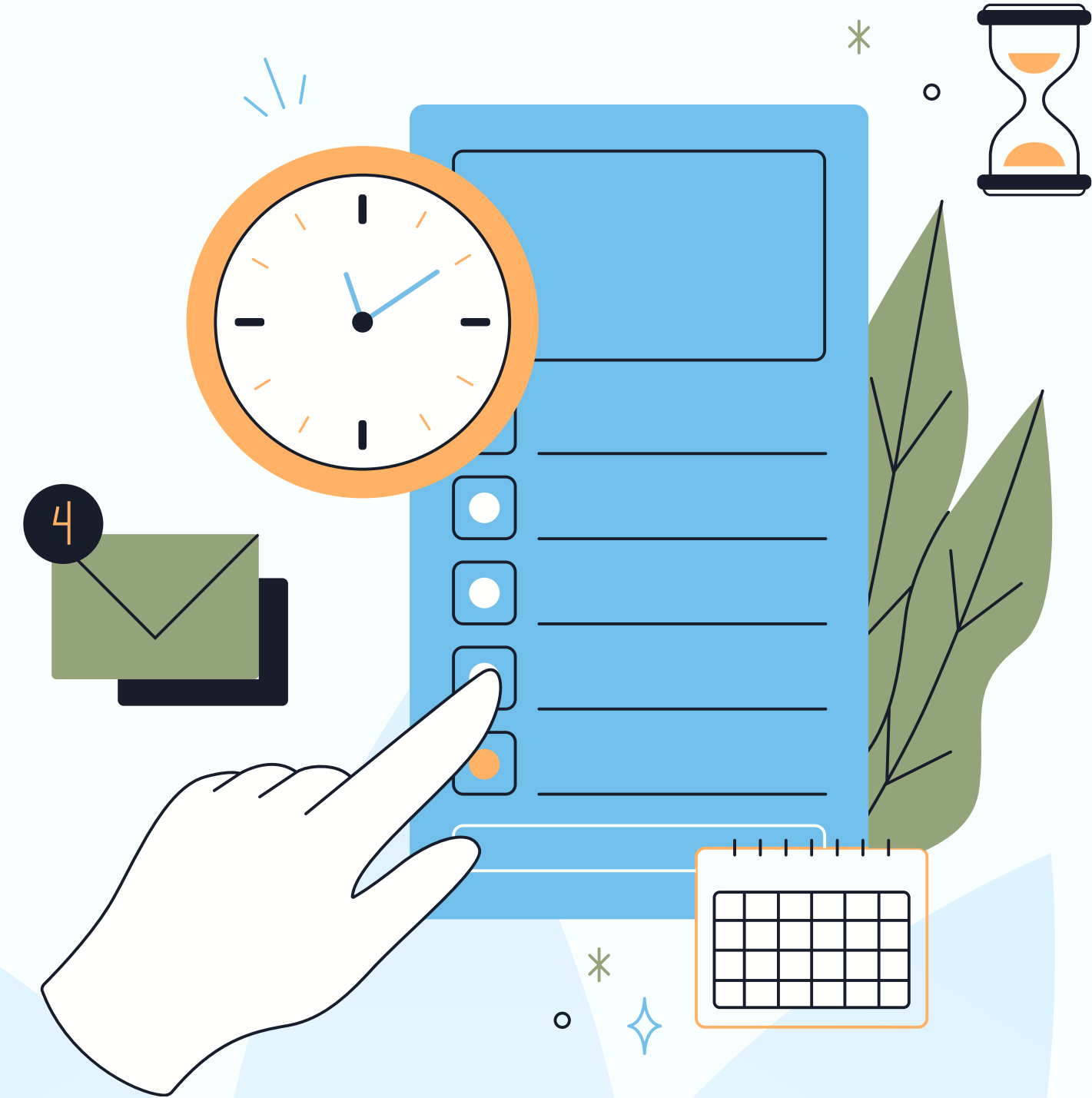


MM-BM — Backdoor Mitigation

Observation: backdoor neurons activate too strongly
The neurons that encode the trigger pattern fire with abnormally large values when any input is presented — not just triggered ones.

Solution: add a learnable upper bound (clamp) per neuron
Instead of retraining, add a clamp parameter z to each neuron:
 $\text{output} = \min(\text{activation}, z)$. Optimize z values on a small set of ~20 clean images so that clean accuracy is preserved but the neuron can never fire above z .

Result: trigger no longer fires
The backdoor neurons' maximum possible activation is now limited. When a backdoored input arrives, the trigger can no longer push the target class logit high enough to override the correct prediction.



Final Results & Comparison

Model	Attack	Train ACC	ASR (before)	Detected?	Pred Target	P-value	ACC (after)	ASR (after)	Mitigation
Model 0	plane→car	88.4%	99.9%	⚠ YES	car (✓)	5.31e-10	86.4%	7.7%	⚠ Partial
Model 1	car→bird	89.1%	97.7%	⚠ YES	bird (✓)	1.57e-12	86.4%	0.8%	⚠ Partial
Model 2	bird→cat	87.9%	98.7%	❌ NO	cat (✓)	3.22e-01	87.3%	20.8%	⚠ Partial

Comparison with Original Paper

Metric	Paper	My Implementation
Dataset	CIFAR-10	CIFAR-10
Architecture	ResNet-18	ResNet-18
ASR Before	~99%	98.77%
ASR After	~3–10%	9.77%
Detection	~90%	2/3 detected



Limitations & Future Work



Current Limitations of My implementation

- Only 3 models were trained
- Only 20 training epochs used
- Only BadNet trigger evaluated
- One detection failure occurred
- Limited computational budget

Limitations & Future Work

Future Research Directions

1. Hybrid Backdoor Detection

Combine:

- Maximum Margin Statistics
- Activation Clustering
- Frequency Analysis
- Entropy-Based Detection

for stronger robustness.

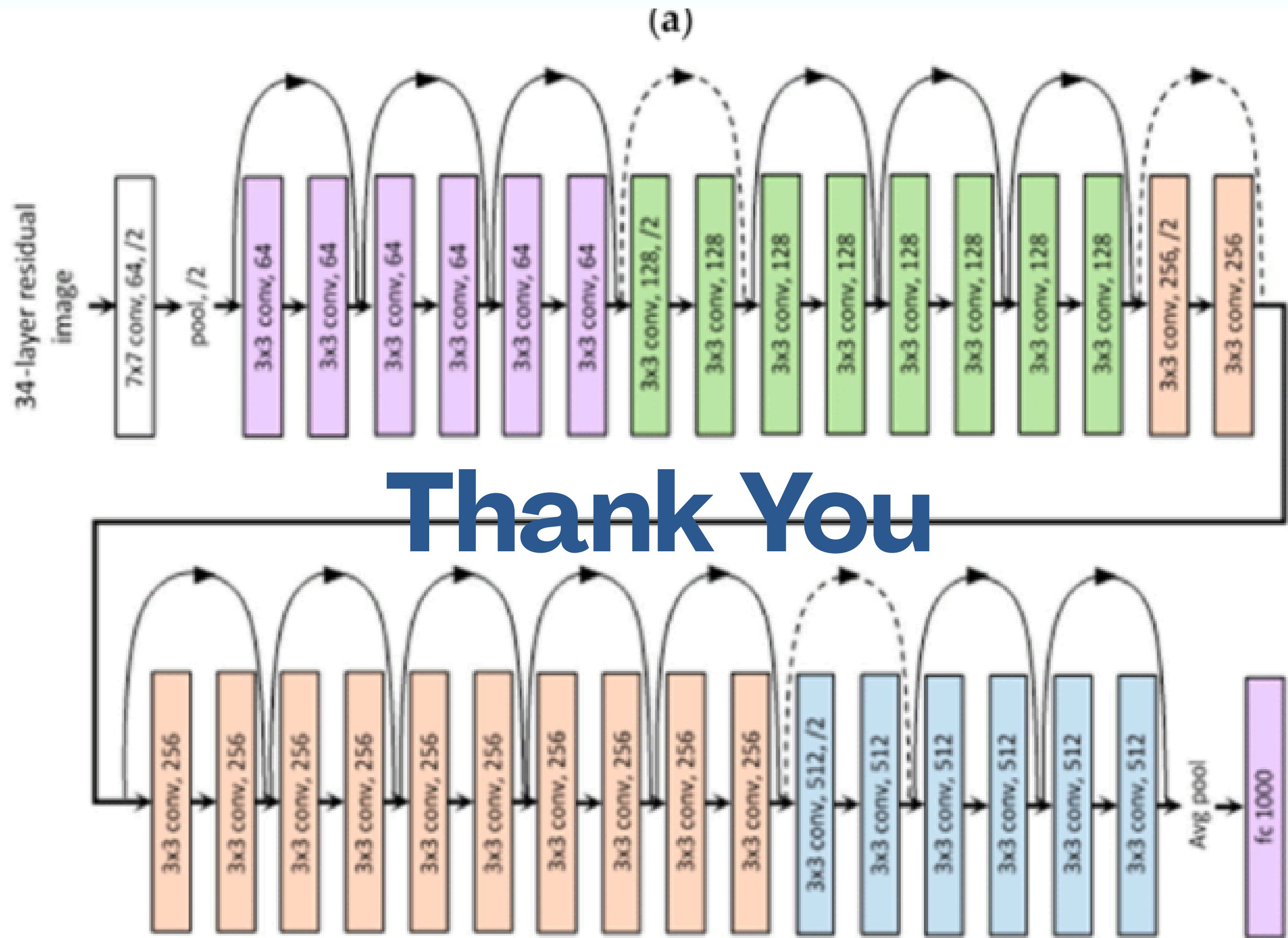
2. Online Backdoor Detection

Instead of detecting after training:
track margin statistics during training
itself.

Advantages:

- early detection
- lower computational cost
- prevention before deployment





Thank You